A Detailed Solution to the Squarefree Divisor Graph Problem

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Abstract

We solve Problem 9 from the curated list of Erdős-style problems. For each positive integer N, let D_N be the graph on $[N] = \{1, 2, \ldots, N\}$ in which two distinct vertices $a < b$ are adjacent precisely when $a \mid b$ and b/a is squarefree. The proposed question asks whether the chromatic number satisfies

$$\chi(D_N) = \Theta\left(\frac{\log N}{\log\log N}\right).$$

We prove the stronger exact formula

$$\chi(D_N) = 1 + \max\{t : p_1 p_2 \cdots p_t \leq N\},$$

where p_i denotes the ith prime. Equivalently, $\chi(D_N) = 1 + K(N)$, where $K(N)$ is the largest possible number of distinct prime divisors of a squarefree integer at most N. The upper bound is given by the coloring $n \mapsto \Omega(n) \pmod{K(N) + 1}$, where $\Omega(n)$ counts prime factors with multiplicity.

1 The problem statement

Throughout this note, $[N]$ means $\{1, 2, \ldots, N\}$. A positive integer m is called *squarefree* if no square p^2 of a prime divides m.

Problem 9: Squarefree divisor graph. *Let D_N be the graph on $[N]$ in which $a < b$ are adjacent if*

$$a \mid b \qquad \text{and} \qquad \frac{b}{a} \text{ is squarefree.}$$

Let

$$\chi(N) = \chi(D_N).$$

Is

$$\chi(N) = \Theta\left(\frac{\log N}{\log\log N}\right)?$$

The lower bound suggested in the original problem is the primorial clique

$$1, \quad p_1, \quad p_1 p_2, \quad \ldots, \quad p_1 p_2 \cdots p_t,$$

where p_i is the ith prime. Any earlier term divides any later term, and the quotient is a product of distinct primes, hence squarefree. Thus these vertices form a clique whenever $p_1 p_2 \cdots p_t \leq N$.

The main point of this note is that this elementary lower bound is exactly sharp.

2 Notation and the exact parameter

For a positive integer n, write

$$\omega(n) = \#\{p : p \text{ is prime and } p \mid n\}$$

for the number of *distinct* prime divisors of n, and write

$$\Omega(n) = \sum_{p^e \| n} e$$

for the number of prime divisors of n counted *with multiplicity*. For example,

$$72 = 2^3 \cdot 3^2, \qquad \omega(72) = 2, \qquad \Omega(72) = 5.$$

If q is squarefree, then every prime exponent in q is 0 or 1, so

$$\Omega(q) = \omega(q).$$

Define

$$K(N) = \max\{\omega(q) : 1 \leq q \leq N,\ q \text{ squarefree}\}.$$

This is the largest number of distinct prime factors that a squarefree integer at most N can have. Equivalently, if

$$P_t = p_1 p_2 \cdots p_t$$

is the tth primorial, with the convention $P_0 = 1$, then

$$K(N) = \max\{t : P_t \leq N\}.$$

Indeed, among squarefree integers with exactly t distinct prime divisors, the smallest is the product of the first t primes. Thus a squarefree integer with t distinct prime factors exists below N exactly when $p_1 p_2 \cdots p_t \leq N$.

3 The exact theorem

Theorem 1. *For every integer $N \geq 1$,*

$$\boxed{\chi(D_N) = K(N) + 1.}$$

Equivalently,

$$\boxed{\chi(D_N) = 1 + \max\{t : p_1 p_2 \cdots p_t \leq N\}.}$$

The proof has two parts. First we build a clique of size $K(N) + 1$, giving the lower bound. Then we give an explicit coloring with exactly $K(N) + 1$ colors, giving the matching upper bound.

4 Lower bound: the primorial clique

Proposition 2. *For every $N \geq 1$,*

$$\chi(D_N) \geq K(N) + 1.$$

Proof. Let

$$K = K(N).$$

By the definition of K, we have

$$p_1 p_2 \cdots p_K \leq N.$$

For $0 \leq i \leq K$, define

$$v_i = p_1 p_2 \cdots p_i,$$

where the empty product is interpreted as $v_0 = 1$.

Thus the vertices are

$$v_0 = 1, \qquad v_1 = p_1, \qquad v_2 = p_1 p_2, \qquad \ldots, \qquad v_K = p_1 p_2 \cdots p_K.$$

Every one of these numbers lies in $[N]$, because

$$v_i \leq v_K = p_1 p_2 \cdots p_K \leq N.$$

We claim that these $K+1$ vertices form a clique in D_N.

Take two different indices $i < j$. Then

$$v_i = p_1 p_2 \cdots p_i$$

divides

$$v_j = p_1 p_2 \cdots p_j.$$

Moreover,

$$\frac{v_j}{v_i} = p_{i+1} p_{i+2} \cdots p_j.$$

This quotient is a product of distinct primes, so it is squarefree. Therefore v_i and v_j are adjacent in D_N.

Since every pair among

$$\{v_0, v_1, \ldots, v_K\}$$

is adjacent, this set is a clique of size $K+1$. The chromatic number of a graph is at least its clique number, because all vertices of a clique must receive different colors in any proper coloring. Hence

$$\chi(D_N) \geq K + 1.$$

□

5 Upper bound: coloring by the total number of prime factors

The key idea is to color an integer n according to $\Omega(n)$ modulo $K(N)+1$. The use of Ω rather than ω is important: if $b = aq$, the quotient q may share primes with a, so the number of distinct prime factors need not behave additively. By contrast, Ω is completely additive.

Lemma 3. *For all positive integers x and y,*

$$\Omega(xy) = \Omega(x) + \Omega(y).$$

Proof. Write the prime factorizations as

$$x = \prod_p p^{\alpha_p}, \qquad y = \prod_p p^{\beta_p},$$

where all but finitely many α_p and β_p are zero. Then

$$xy = \prod_p p^{\alpha_p+\beta_p}.$$

Therefore

$$\Omega(xy) = \sum_p (\alpha_p + \beta_p) = \sum_p \alpha_p + \sum_p \beta_p = \Omega(x) + \Omega(y).$$

□

Proposition 4. *For every $N \geq 1$,*

$$\chi(D_N) \leq K(N) + 1.$$

Proof. Again put

$$K = K(N).$$

We define a coloring

$$c : [N] \longrightarrow \{0, 1, 2, \ldots, K\}$$

by

$$c(n) \equiv \Omega(n) \pmod{K+1}.$$

In other words, two integers receive the same color precisely when their values of Ω are congruent modulo $K+1$.

We prove that this is a proper coloring of D_N. Let $a < b$ be adjacent vertices in D_N. By the definition of the graph, there is a squarefree integer $q > 1$ such that

$$b = aq.$$

Since $b \leq N$ and $a \geq 1$, we also have

$$q = \frac{b}{a} \leq N.$$

Because q is squarefree and at most N, the definition of $K(N)$ gives

$$\omega(q) \leq K.$$

Because $q > 1$, it has at least one prime divisor, so

$$\omega(q) \geq 1.$$

Thus

$$1 \leq \omega(q) \leq K.$$

Since q is squarefree,

$$\Omega(q) = \omega(q).$$

Using the complete additivity of Ω, Lemma 3 gives

$$\Omega(b) = \Omega(aq) = \Omega(a) + \Omega(q) = \Omega(a) + \omega(q).$$

Therefore

$$\Omega(b) - \Omega(a) = \omega(q).$$

But $\omega(q)$ is one of the integers

$$1, 2, \ldots, K.$$

In particular,

$$\omega(q) \not\equiv 0 \pmod{K+1}.$$

Hence

$$\Omega(b) \not\equiv \Omega(a) \pmod{K+1},$$

which means

$$c(b) \neq c(a).$$

Thus the endpoints of every edge receive different colors. So c is a proper coloring using $K + 1$ colors, and consequently

$$\chi(D_N) \leq K + 1.$$

$\square$

Proof of Theorem 1. Proposition 2 gives

$$\chi(D_N) \geq K(N) + 1.$$

Proposition 4 gives

$$\chi(D_N) \leq K(N) + 1.$$

The two inequalities match, so

$$\chi(D_N) = K(N) + 1.$$

Finally, as explained above, $K(N)$ is equivalently the largest t for which

$$p_1 p_2 \cdots p_t \leq N.$$

This proves the exact formula. $\square$

6 Why the coloring works despite shared primes

It is worth spelling out the only possible subtlety in the upper bound. Suppose $a < b$ are adjacent and $b = aq$ with q squarefree. The quotient q need not be coprime to a.

For example, take

$$a = 12 = 2^2 \cdot 3, \qquad q = 6 = 2 \cdot 3, \qquad b = 72 = 2^3 \cdot 3^2.$$

Then q is squarefree, so a and b would be adjacent if both lie in $[N]$. But

$$\omega(a) = 2, \qquad \omega(q) = 2, \qquad \omega(b) = 2,$$

so $\omega(b) - \omega(a)$ is not equal to $\omega(q)$. Thus coloring by $\omega(n)$ would not have the simple additive behavior needed for the proof.

By contrast,

$$\Omega(a) = 3, \qquad \Omega(q) = 2, \qquad \Omega(b) = 5,$$

and indeed

$$\Omega(b) - \Omega(a) = 2 = \Omega(q) = \omega(q).$$

This is exactly why the proof colors by $\Omega(n)$ rather than by $\omega(n)$.

7 Asymptotic consequence

The exact formula immediately implies the asymptotic estimate asked for in the problem. We include the standard argument for completeness.

Theorem 5. *As $N \to \infty$,*

$$\chi(D_N) = (1 + o(1)) \frac{\log N}{\log \log N}.$$

In particular,

$$\chi(D_N) = \Theta\left(\frac{\log N}{\log \log N}\right).$$

Proof. By Theorem 1,

$$\chi(D_N) = K(N) + 1,$$

where

$$K(N) = \max\{t : p_1 p_2 \cdots p_t \le N\}.$$

Thus it remains to estimate the largest t such that the tth primorial is at most N.

Let

$$P_t = p_1 p_2 \cdots p_t.$$

Taking logarithms gives

$$\log P_t = \sum_{i=1}^{t} \log p_i.$$

This sum is the Chebyshev function evaluated at p_t:

$$\log P_t = \vartheta(p_t),$$

where
$$\vartheta(x) = \sum_{p \le x} \log p.$$
By the prime number theorem in the form
$$\vartheta(x) \sim x,$$
and the standard consequence
$$p_t \sim t \log t,$$
we obtain
$$\log P_t = \vartheta(p_t) \sim p_t \sim t \log t.$$
So
$$\log P_t = (1 + o(1)) t \log t.$$
Now $K(N)$ is the largest t for which $P_t \le N$, equivalently the largest t for which
$$\log P_t \le \log N.$$
Since
$$\log P_t = (1 + o(1)) t \log t,$$
inverting the relation gives
$$K(N) = (1 + o(1)) \frac{\log N}{\log \log N}.$$
For clarity, here is the inversion explicitly. Put
$$L = \log N.$$
If
$$t = (1 - \varepsilon) \frac{L}{\log L},$$
then
$$\log t = \log L - \log \log L + O_\varepsilon(1) = (1 + o(1)) \log L,$$
so
$$t \log t = (1 - \varepsilon + o(1)) L.$$
Hence $P_t \le N$ for all sufficiently large N, giving
$$K(N) \ge (1 - \varepsilon) \frac{\log N}{\log \log N}$$
for large N. Similarly, if
$$t = (1 + \varepsilon) \frac{L}{\log L},$$
then
$$t \log t = (1 + \varepsilon + o(1)) L,$$
so $P_t > N$ for all sufficiently large N, giving
$$K(N) < (1 + \varepsilon) \frac{\log N}{\log \log N}$$

for large N. Since $\varepsilon > 0$ is arbitrary,

$$K(N) = (1 + o(1)) \frac{\log N}{\log \log N}.$$

Adding 1 does not affect the asymptotic, so

$$\chi(D_N) = K(N) + 1 = (1 + o(1)) \frac{\log N}{\log \log N}.$$

□

8 Small examples

The exact formula also makes small cases transparent.

- If $N = 1$, then $K(1) = 0$ and D_1 has one vertex, so $\chi(D_1) = 1 = K(1) + 1$.
- If $2 \leq N < 6$, then $K(N) = 1$, since the smallest squarefree integer with two prime factors is $2 \cdot 3 = 6$. Hence $\chi(D_N) = 2$.
- If $6 \leq N < 30$, then $K(N) = 2$, since $2 \cdot 3 \leq N < 2 \cdot 3 \cdot 5$. Hence $\chi(D_N) = 3$.
- If $30 \leq N < 210$, then $K(N) = 3$, and $\chi(D_N) = 4$.

For instance, when $6 \leq N < 30$, the clique

$$1, \quad 2, \quad 6$$

forces three colors, and the coloring

$$n \longmapsto \Omega(n) \pmod 3$$

uses exactly three colors and is proper.

9 Conclusion

The proposed estimate is true, and the chromatic number is exactly controlled by the largest primorial below N:

$$\boxed{\chi(D_N) = 1 + \max\{t : p_1 p_2 \cdots p_t \leq N\}.}$$

The lower bound is the obvious chain of initial primorials. The matching upper bound comes from the fact that every edge $a \sim b$ has quotient $q = b/a$ squarefree, so the difference

$$\Omega(b) - \Omega(a)$$

is exactly the number of prime factors of q, an integer between 1 and $K(N)$. Therefore this difference is never 0 modulo $K(N) + 1$, which makes

$$n \mapsto \Omega(n) \pmod{K(N) + 1}$$

a proper coloring.